



Jan Sobczyk

NUWRO

PAST AND FUTURE



NuWro Workshop, 16-18 September, 2026



Welcome
to Wrocław!!!

It is the second
NuWro Workshop...

NuWro Workshop 2017

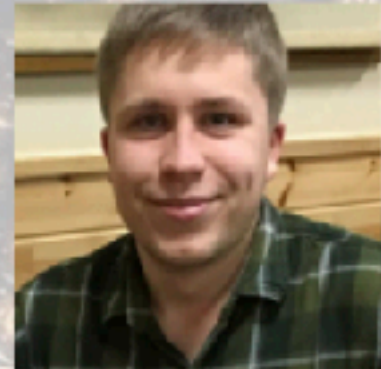
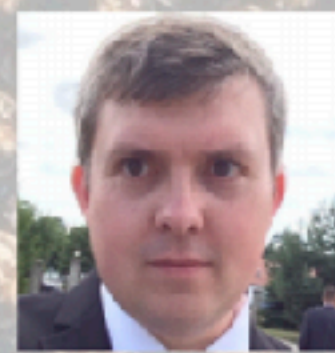


NuWro team

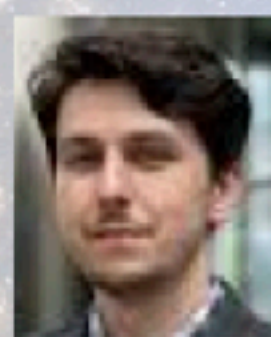
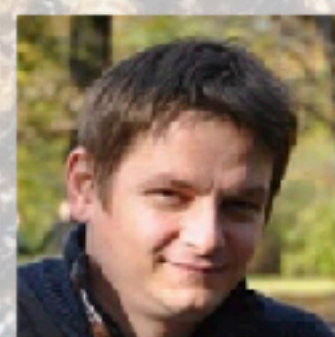
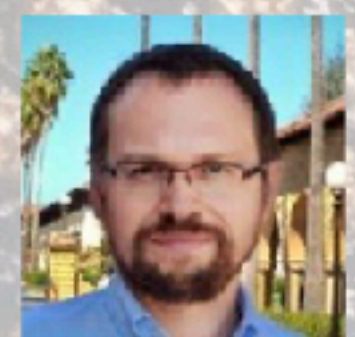
Two main authors of NuWro: Cezary Juszczak (structure of the code) and Jan Sobczyk (physics)



A major part of NuWro physics models were implemented by PhD students, from left: Jarosław Nowak, Tomasz Golan, Kajetan Niewczas, Qiyu Yan, Rwik Dharmapal Banerjee, Hemant Prasad



Important contributions from left: Artur Ankowski, Tomasz Bonus, Krzysztof Graczyk, Michał Siemaszko, Chris Thorpe, Dmitry Zhuridov, Jakub Żmuda



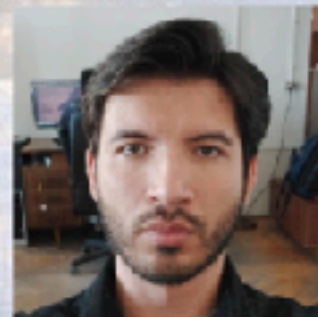
Rewighting tools added by Patrick Stowell and Luke Pickering



Inspiration - Danka Kiełczewska



NuWro AI studies: Luis Bonilla, Beata Kowal



2026 - **difficult** transition time:
Rwik Dharmapal Benerjee and Hemant Prasad are completing their thesis, Luis Bonilla will also leave Wrocław soon.

Hopefully, two new members: PhD students in WEAVE-UNISONO NCN grant and a new master student



The NuWro Timeline

WROCLAW NEUTRINO EVENT GENERATOR · 2003 - 2027

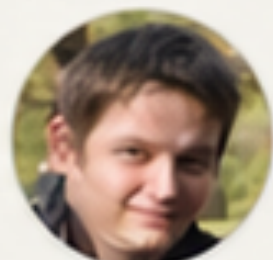
INSTITUTE OF THEORETICAL PHYSICS
UNIVERSITY OF WROCLAW



2003-04

Danka Kielczewska

Discussions that started it



2005

Krzysztof M. Graczyk

WroNG - single pion production with Sobczyk & Nowak



2007

The name NuWro is first used

2012

Jan Sobczyk

Hadronic model for MEC



2014

Tomek Golan

Thesis



2016

Patrick Stowell · Luke Pickering

Rewighting tools



2019

Christopher Thorpe

Hyperon production



2023

Kajetan Niewczas

Thesis



2026-27

Hemant Prasad

Thesis in preparation

ν

2005-06

Czarek Juszczak

The core of NuWro is designed



2006

Jarek Nowak

Thesis



2010

Artur Ankowski

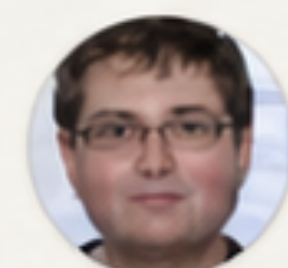
Spectral function with C. Juszczak



2012

Jakub Żmuda

Valencia MEC model



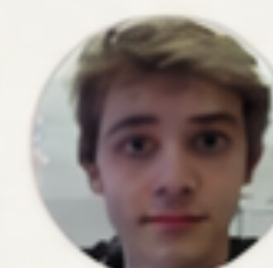
2015

eWro - electron scattering

2019

Tomasz Bonus

MEC scaling



2020

Dmitry Zhuridov

Scattering on electrons



2024

Qiyu Yan

Ghent pion production model



2026-27

Rwik Dharmapal Banerjee

Thesis in preparation



Eighteen milestones - larger marks and larger portraits are major ones

Two theses still in preparation

Goals of the Workshop

Although the workshop bears the NuWro name, its scope will be broader, focusing on **general aspects of modeling neutrino interactions**. The reference to NuWro in the title reflects a **broader goal of the neutrino interaction community: the development of improved models suitable for implementation in Monte Carlo generators, thereby supporting the success of current and future neutrino oscillation experiments**. At the same time, the workshop will provide an excellent opportunity to present **recent efforts to enhance the quality of the physics models implemented in NuWro**.

Goals of the Workshop

Specifically NuWro goals:

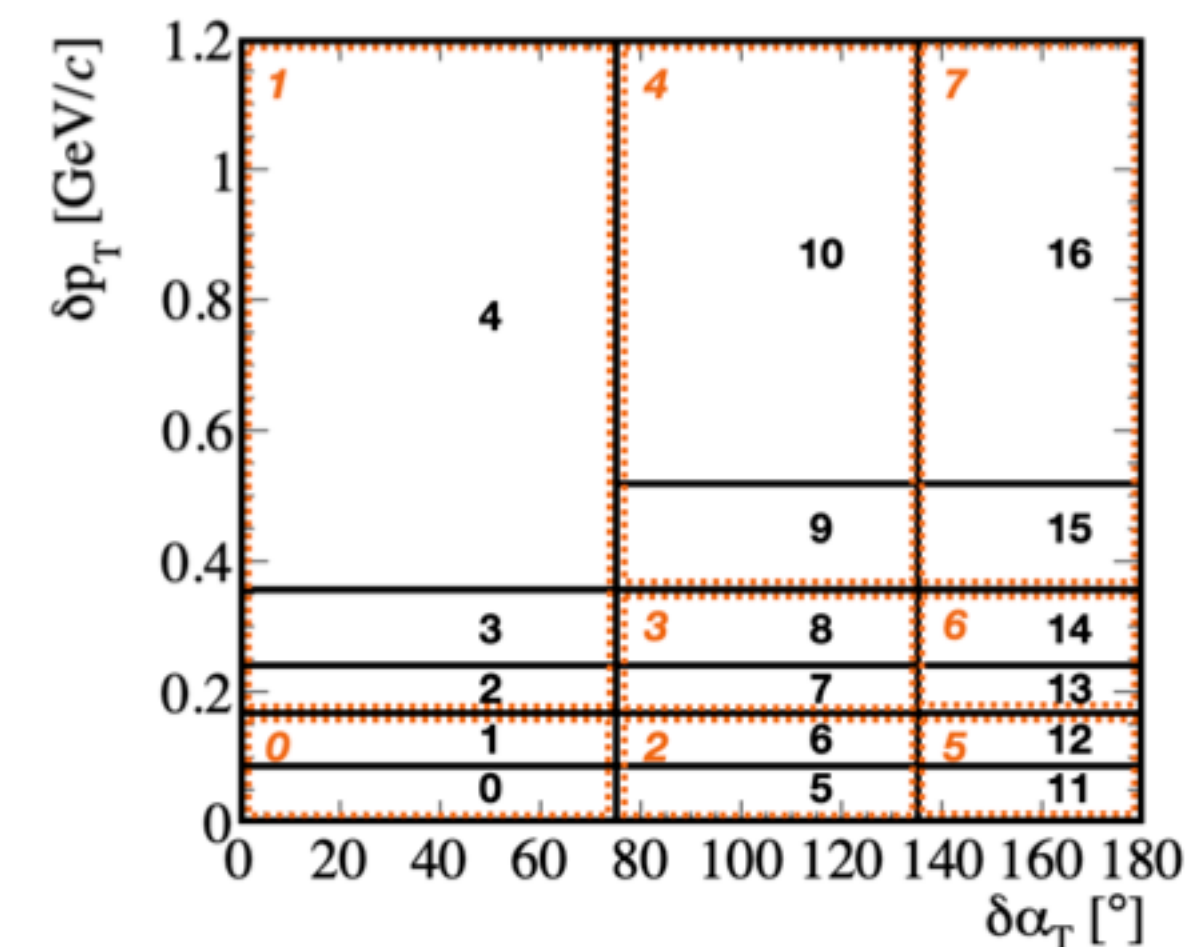
- NuWro versions 25.03 and later are significantly different compared to 21.09.
- New version(s) must be critically benchmarked.
- It would be great to formulate a list of priorities for future developments.

NuWro challenges

Studies employing NuWro 25.03 or 25.11

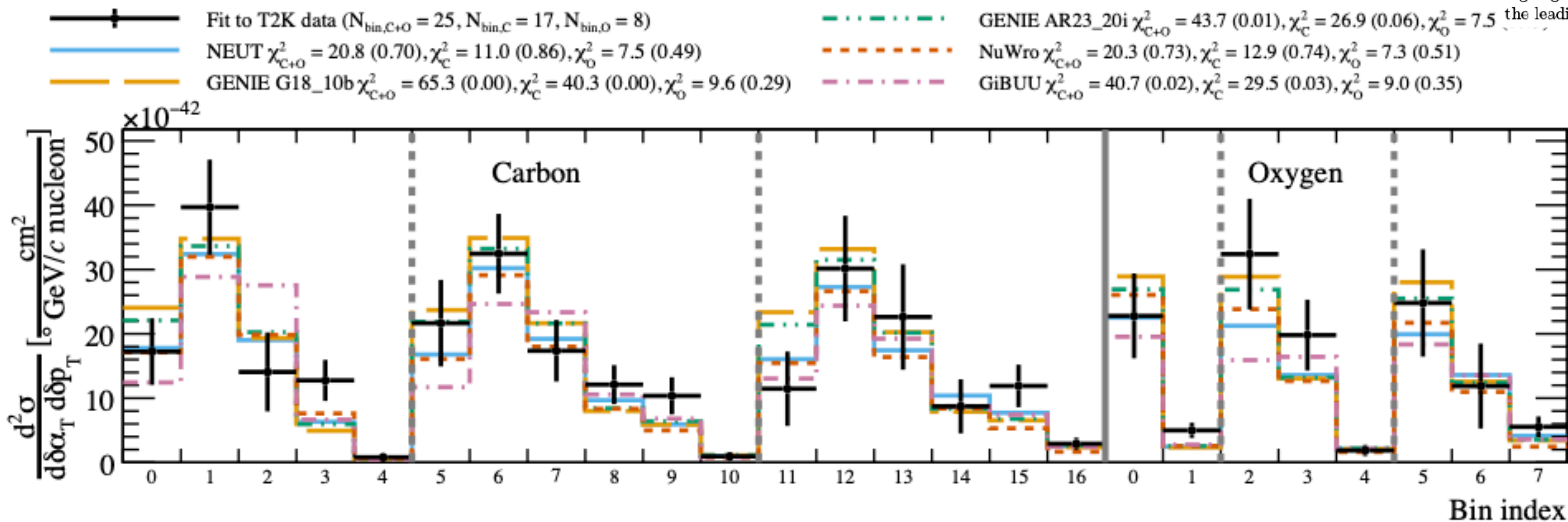
T2K, arXiv 2607.10638v2, TKI, carbon, oxygen targets

signal is CC0 π



(a) $\delta p_T - \delta \alpha_T$

These constraints are defined as $0.225 < p_\mu < 10$ GeV/c and $\cos \theta_\mu > -0.6$ for the muon momentum and scattering angle, and $0.525 < p_p < 1$ GeV/c and $\cos \theta_p > 0.3$ for the leading proton momentum and scattering angle.



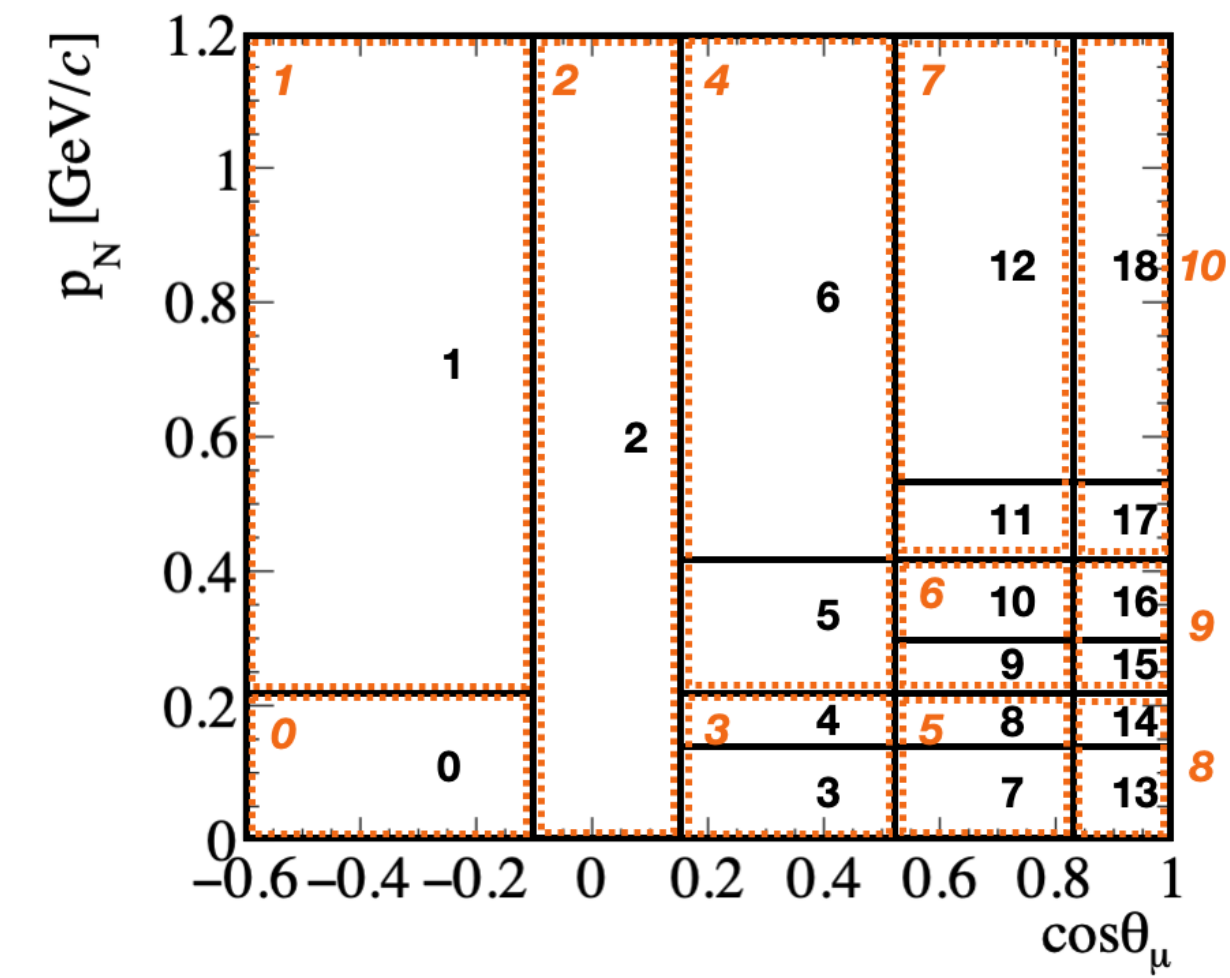
(a) $\delta p_T - \delta \alpha_T$

NuWro challenges

Studies employing NuWro 25.03 or 25.11

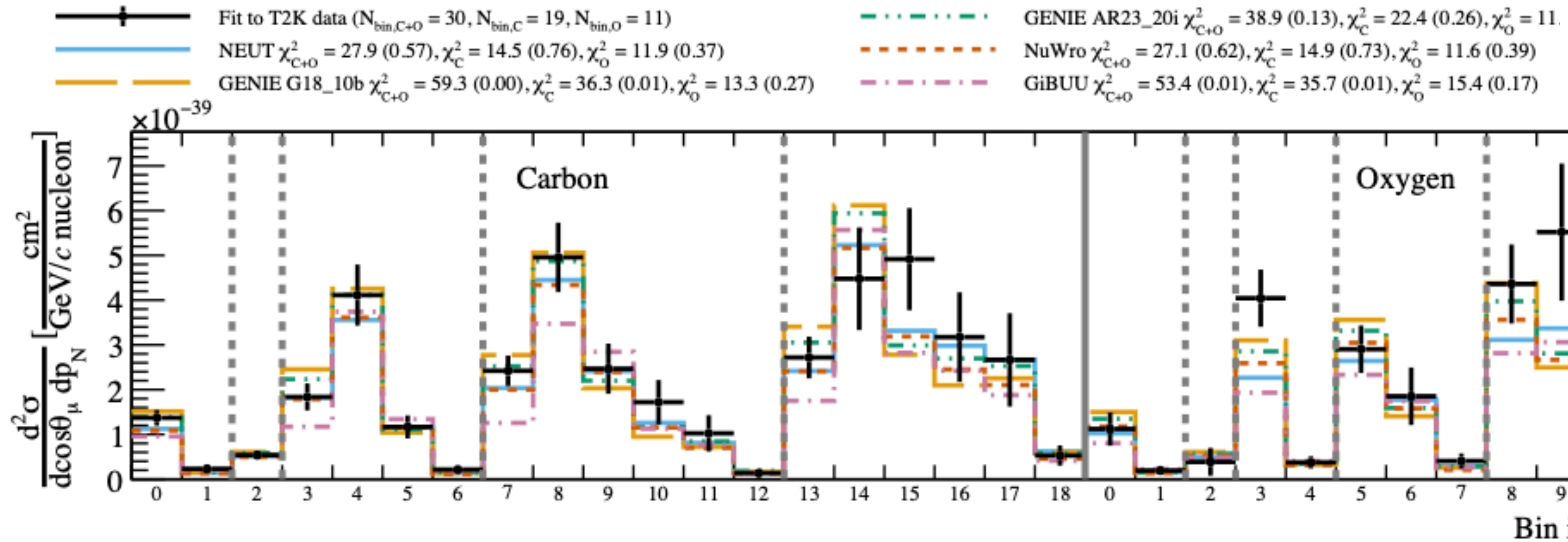
T2K, arXiv 2607.10638v2, TKI, carbon, oxygen targets

signal is $CC0\pi$



(b) $p_N - \cos \theta_\mu$

These constraints are defined as $0.225 < p_\mu < 10$ GeV/c and $\cos \theta_\mu > -0.6$ for the muon momentum and scattering angle, and $0.525 < p_p < 1$ GeV/c and $\cos \theta_p > 0.3$ for the leading proton momentum and scattering angle.



(b) $p_N - \cos \theta_\mu$

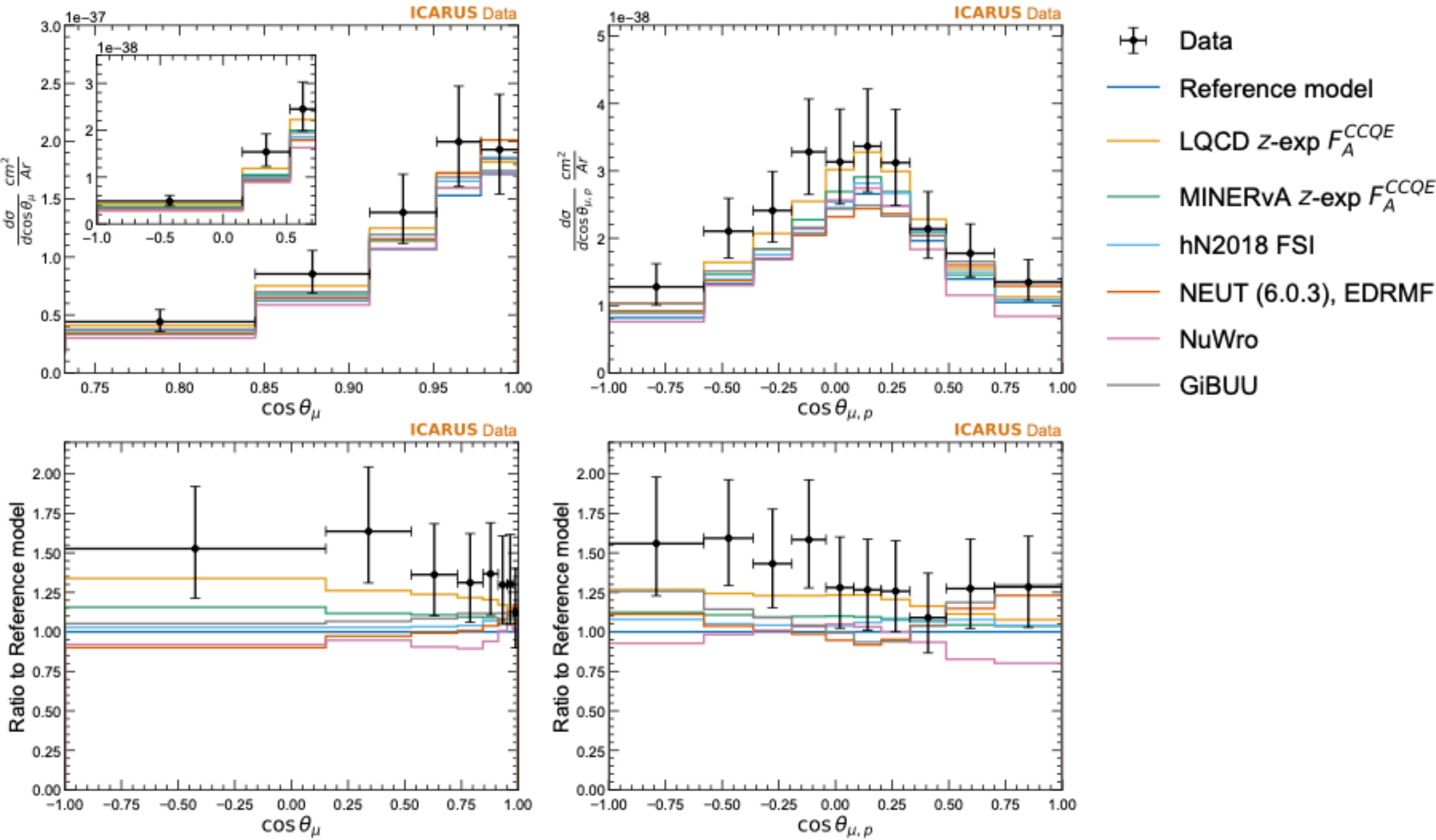
NuWro challenges

Studies employing NuWro 25.03 or 25.11

ICARUS, Phys. Rev. D 114,
052002 (2026), signal is $CC0\pi$

straints are applied in the definition of the signal. Muons are required to have momentum above 226 MeV/c, corresponding to a distance of approximately 50 cm traveled in liquid argon. The leading proton in the final state is required to have momentum between 400 and 1000 MeV/c.

Prediction	Angular variables	
	$\chi^2/ndof$	p-value
Reference model	13.44/18	0.765
LQCD z -exp F_A^{CCQE}	7.27/18	0.988
MINERvA z -exp F_A^{CCQE}	9.40/18	0.950
hN2018 FSI	14.47/18	0.698
NEUT (6.0.3), EDRMF	23.34/18	0.178
NuWro	17.34/18	0.500
GiBUU	15.88/18	0.601



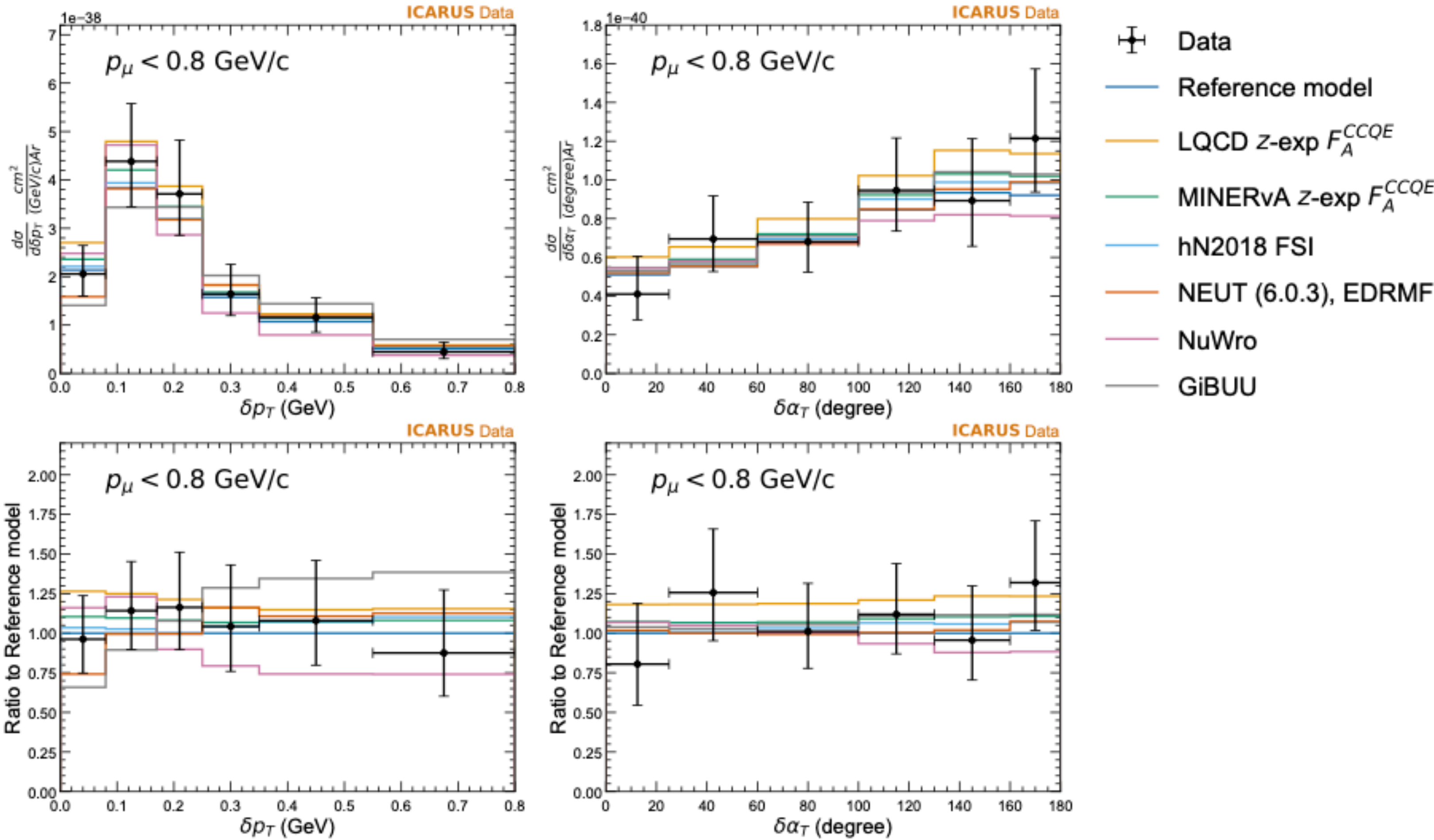
NuWro challenges

Studies employing NuWro 25.03 or 25.11

ICARUS, Phys. Rev. D 114, 052002 (2026), signal is $CC0\pi$

straints are applied in the definition of the signal. Muons are required to have momentum above 226 MeV/c, corresponding to a distance of approximately 50 cm traveled in liquid argon. The leading proton in the final state is required to have momentum between 400 and 1000 MeV/c.

Prediction	TKI variables	
	$\chi^2/ndof$	p-value
Reference model	4.44/12	0.974
LQCD z -exp F_A^{CCQE}	4.58/12	0.971
MINERvA z -exp F_A^{CCQE}	4.15/12	0.981
hN2018 FSI	4.68/12	0.968
NEUT (6.0.3), EDRMF	5.49/12	0.940
NuWro	13.73/12	0.318
GiBUU	13.38/12	0.342



NuWro challenges

Studies employing NuWro 25.03 or 25.11

MicroBooNE, arXiv 2606.13613 , COH CC π^+

TABLE I. Comparison of the measured ν_μ CC π_c^+ cross section on argon with model predictions. All values are flux-averaged and expressed in units of $10^{-40} \text{ cm}^2/\text{Ar}$.

Model/Measurement	Cross Section	χ^2/n_{bins}
Measured (MicroBooNE)	$9.1 \pm 1.2_{\text{stat}} \pm 1.2_{\text{syst}}$	
GENIE 3.0.6 G18 μB	5.0	5.8/1
GENIE 3.6.2 G18_10a_02_11a (RS)	10.1	0.3/1
NEUT 6.1.4	10.7	0.9/1
NuWro 25.11	14.0	8.3/1

NuWro COH model
has not been changed for many years!

NuWro challenges

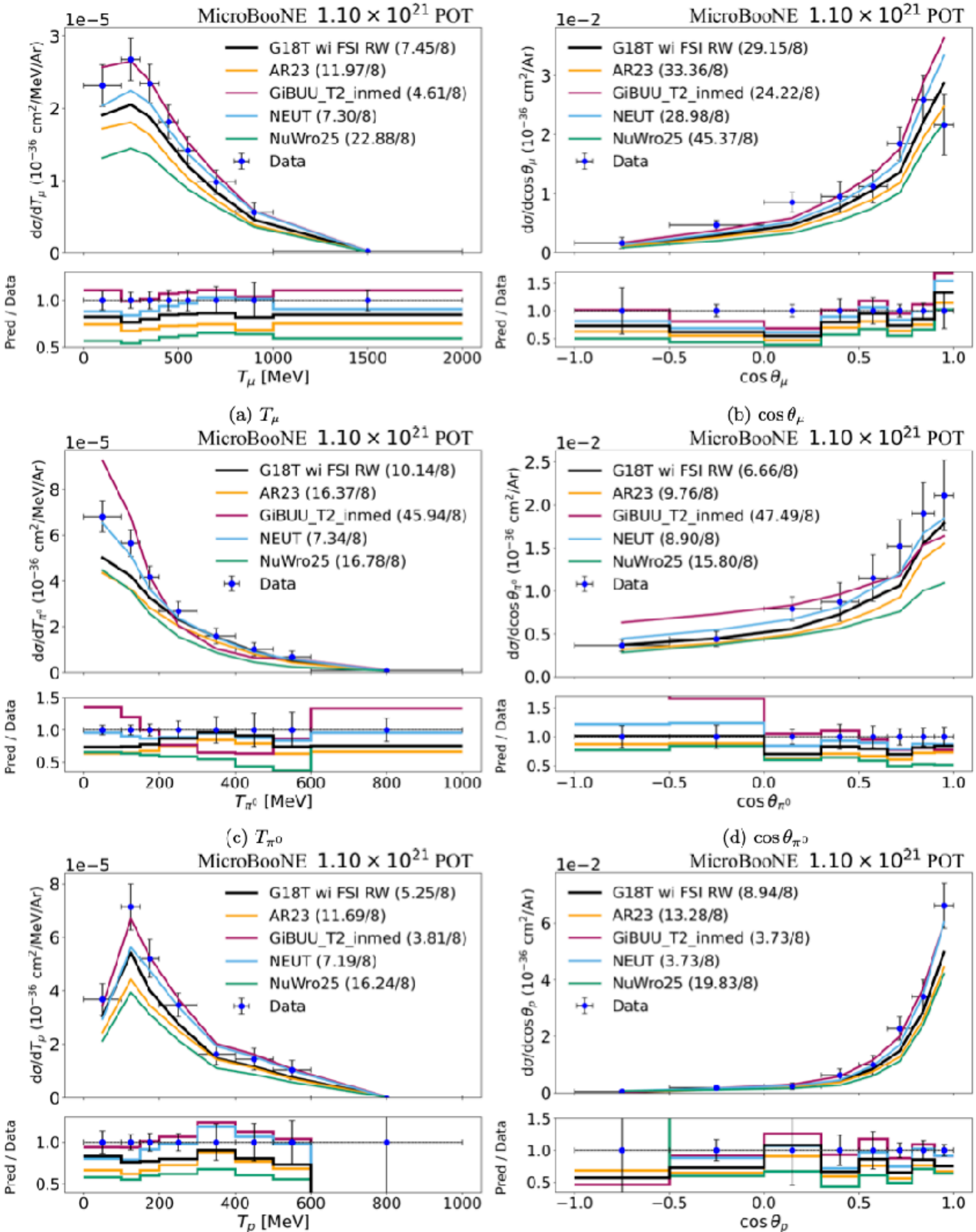
Studies employing NuWro 25.03 or 25.11

MicroBooNE, arXiv 2606.23559, CCNp1 π^0

muon kinetic energy (T_μ) above 40 MeV, at least one proton with T_p above 60 MeV, and no charged pions or heavier mesons with kinetic energy above 40 MeV. The

Observable	ndf	G18T (FSI RW)	AR23	NEUT	NuWro25.11	GiBUU_T2_inmed
T_μ	8	7.45	11.97	7.30	22.88	4.61
$\cos \theta_\mu$	8	29.15	33.36	28.98	45.37	24.22
T_{π^0}	8	10.14	16.37	7.34	16.78	45.94
$\cos \theta_{\pi^0}$	8	6.66	9.76	8.90	15.80	47.49
T_p	8	5.25	11.69	7.19	16.24	3.81
$\cos \theta_p$	8	8.94	13.28	3.73	19.83	3.73

In general, NuWro is below the data
in normalization



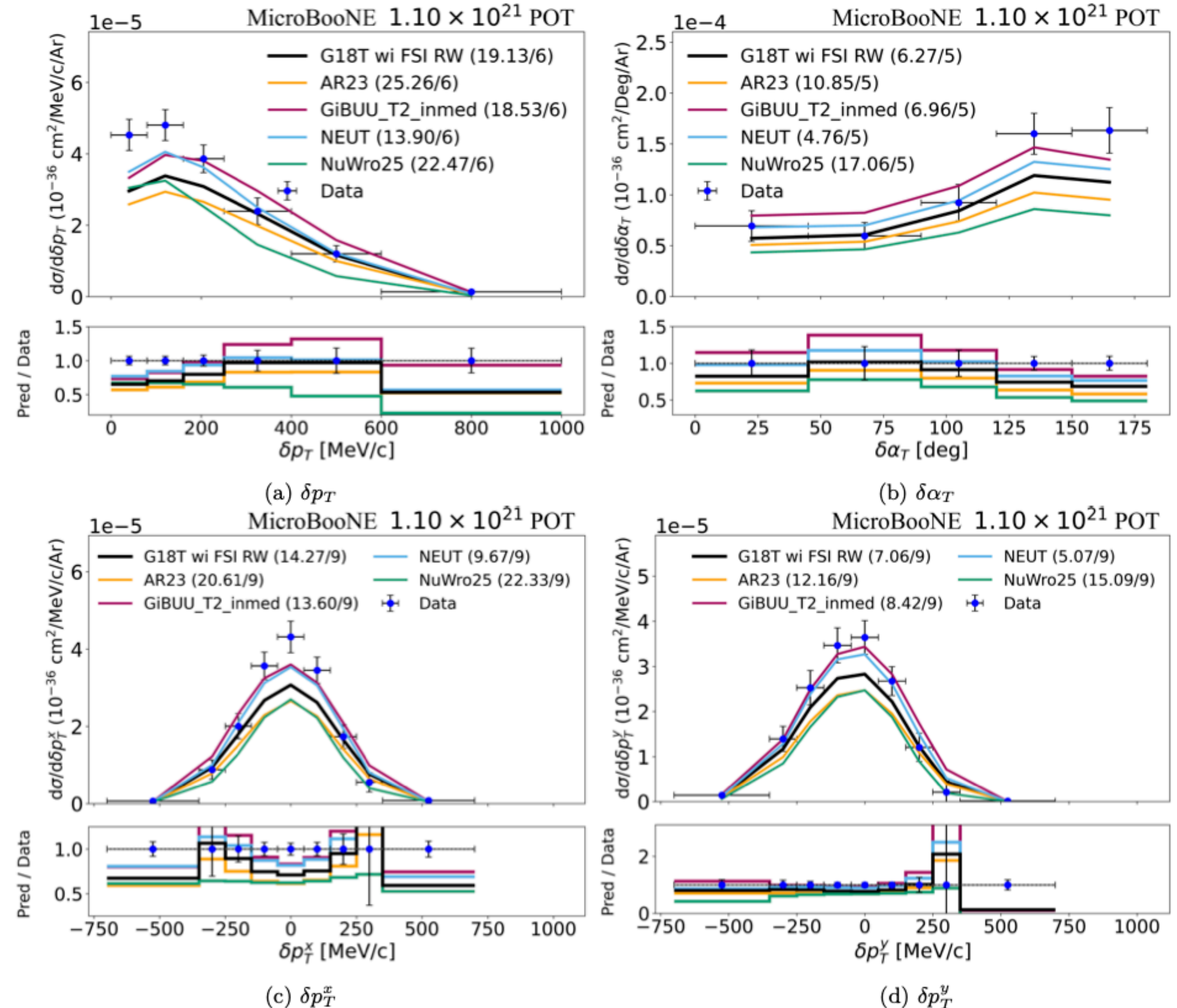
NuWro challenges

Studies employing NuWro 25.03 or 25.04

MicroBooNE, arXiv 2606.23559, CCNp1

muon kinetic energy (T_μ) above 40 MeV, at least one proton with T_p above 60 MeV, and no charged pions or heavier mesons with kinetic energy above 40 MeV. The

Observable	ndf	G18T (FSI RW)	AR23	NEUT	NuWro25.11	GiBUU_T2_inmed
δp_T	6	19.13	25.26	13.90	22.47	18.53
δp_T^x	9	14.27	20.61	9.67	22.33	13.60
δp_T^y	9	7.06	12.16	5.07	15.09	8.42
$\delta \alpha_T$	5	6.27	10.85	4.76	17.06	6.96



NuWro challenges

Generator model	Ground state	QE	2p2h	RPP	SIS and hadronisation	DIS and [128, 141] hadronisation	FSI [173]
NuWro 25	LFG	Ref [249]	Valencia 20 [199, 244]	Hybrid [251]	Use Hybrid [251] for $W < 1.6$ GeV	GRV98 $W > 1.6$ GeV: Pyth.6	NuWro Casc.

Studies employing
NuWro 25.03 or 25.11

S. Dolan₁, L. Pickering₃, P.
Stowell₃,
C. Wilkinson₄, C. Wret₅,
arXiv 2605.28671

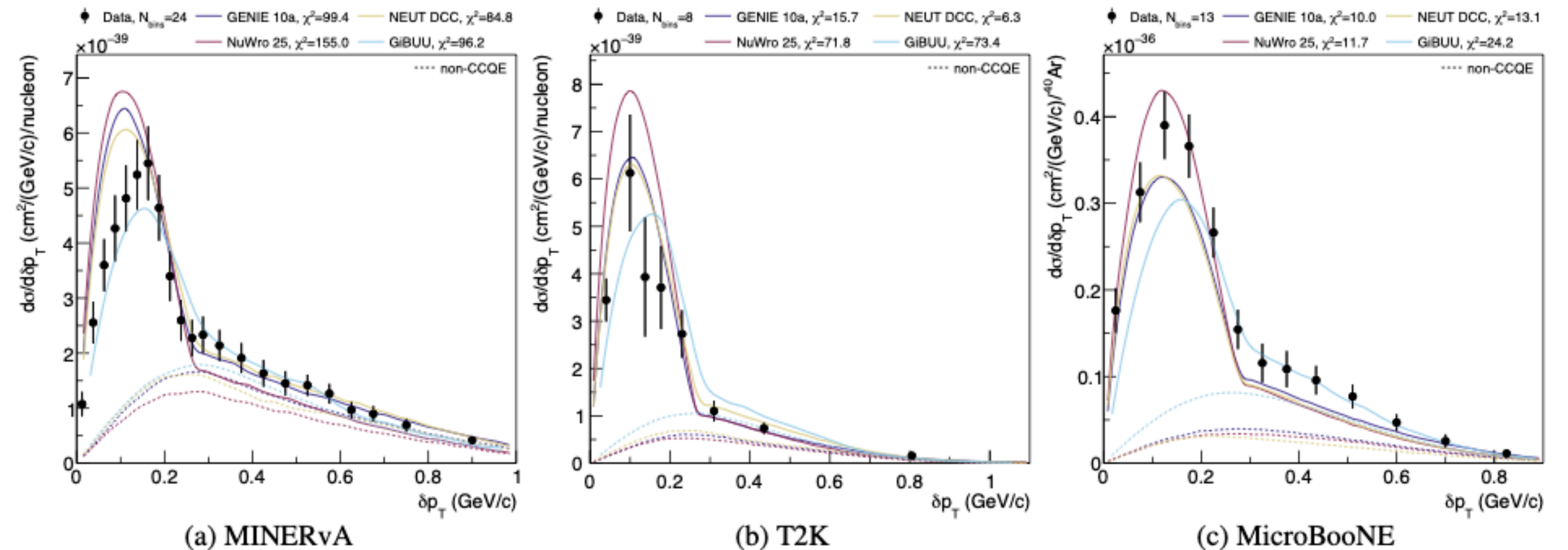


Fig. 18: Comparisons of event generator configurations to measurements of the transverse momentum imbalance in ν_μ CC0 π Np events from MINERvA on C₈H₈ [291], T2K on C₈H₈ [289], and MicroBooNE on ⁴⁰Ar [296]. The χ^2 shown in the legend indicate the level of agreement between the models and measurements and are calculated using the full covariance matrix provided in the experimental data releases.

NuWro challenges

Generator model	Ground state	QE	2p2h	RPP	SIS and hadronisation	DIS and [128, 141] hadronisation	FSI [173]
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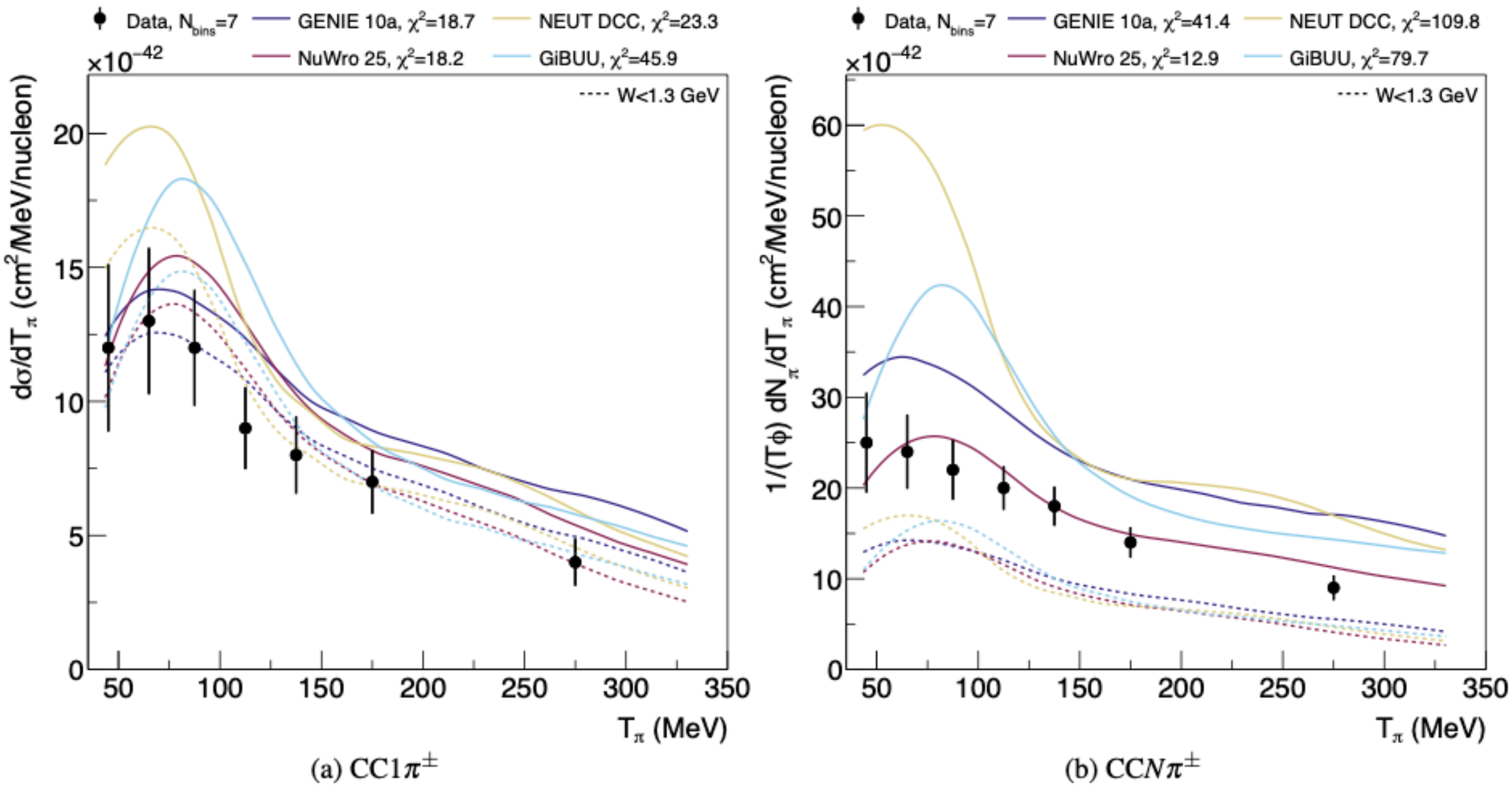


Fig. 20: $\text{CC}1\pi^\pm$ [281] (left) and $\text{CCN}\pi^\pm$ [447] (right) measurements made by MINERvA on a C_8H_8 target compared to current generator models. Dashed lines show the contributions that come primarily from the Delta resonance ($W < 1.3$ GeV). The χ^2 values are shown in the legend. The $\text{CCN}\pi^+$ measurement includes *all* pions in an event, not just the highest momentum pion. The χ^2 shown in the legend indicate the level of agreement between the models and measurements and are calculated using the full covariance matrix provided in the experimental data releases.

NuWro challenges

Generator model	Ground state	QE	2p2h	RPP	SIS and hadronisation	DIS and [128, 141] hadronisation	FSI [173]
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Studies employing
NuWro 25.03 or 25.11

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Stowell₃,
C. Wilkinson₄, C. Wret₅,
arXiv 2605.28671

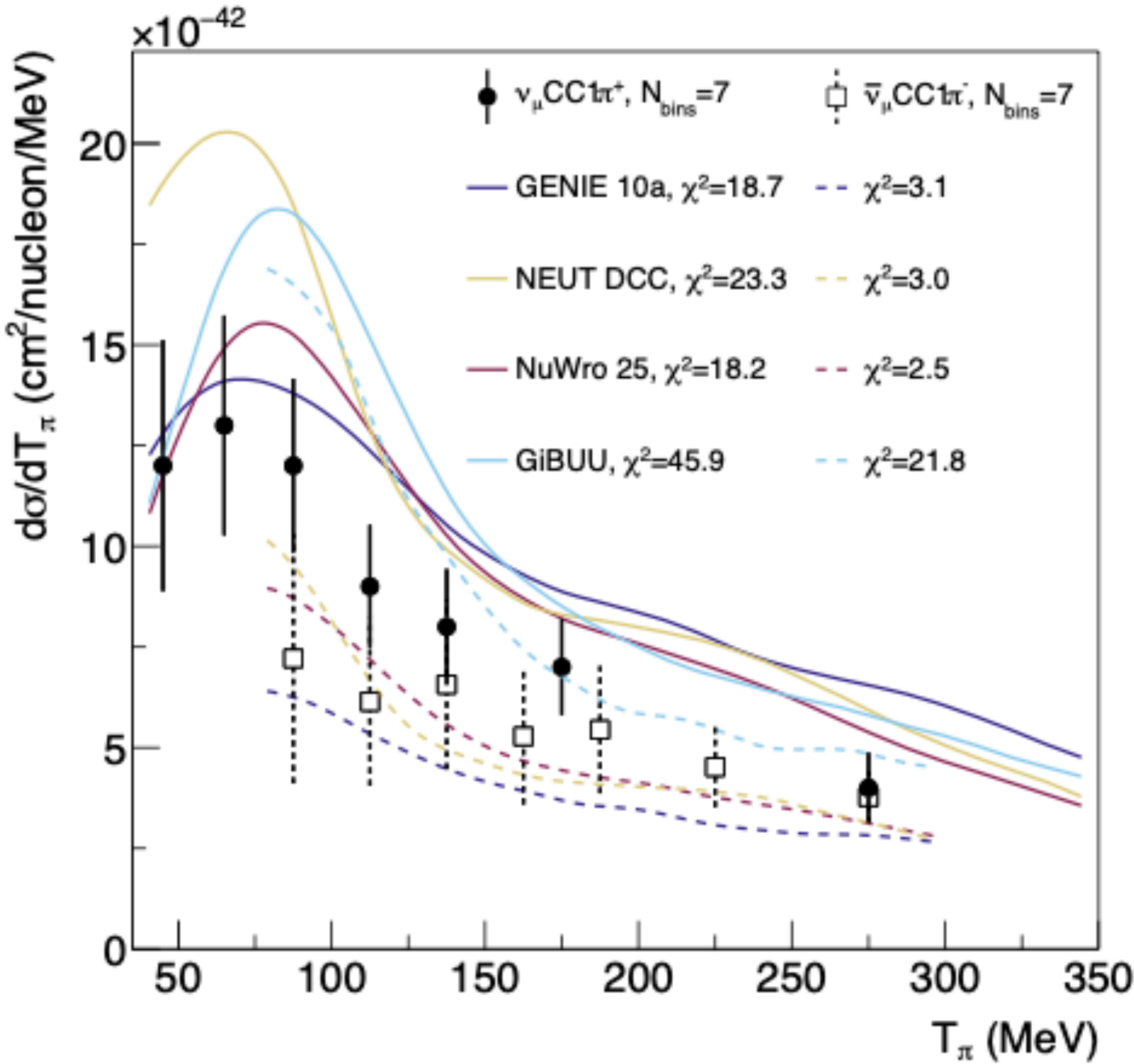


Fig. 21: MINERvA's measurements of the differential cross sections on a C₈H₈ target in pion kinetic energy for $\nu_\mu\text{CC}1\pi^+$ [281] (solid circle, solid lines) and $\bar{\nu}_\mu\text{CC}1\pi^-$ [448] (open square, dashed lines) final states,

NuWro challenges

Generator model	Ground state	QE	2p2h	RPP	SIS and hadronisation	DIS and [128, 141] hadronisation	FSI [173]
NuWro 25	LFG	Ref [249]	Valencia 20 [199, 244]	Hybrid [251]	Use Hybrid [251] for $W < 1.6$ GeV	GRV98 $W > 1.6$ GeV: Pyth.6	NuWro Casc.

Studies employing
NuWro 25.03 or 25.11

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C. Wilkinson₄, C. Wret₅,
arXiv 2605.28671

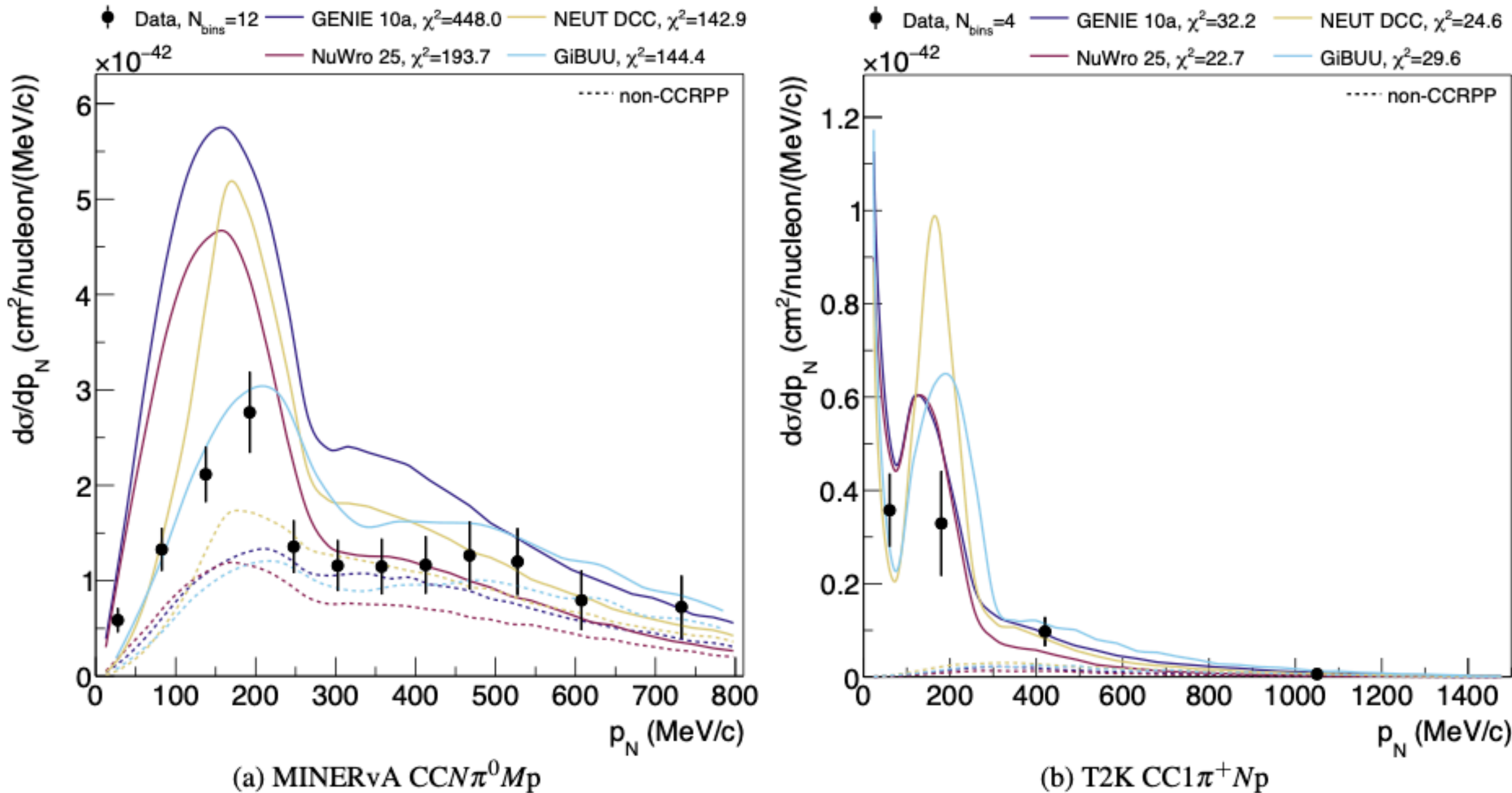


Fig. 22: A measurement of the $CCN\pi^0 Mp$ ($N > 0, M > 0$) (left) cross section from MINERvA on C_8H_8 [293] and the $CC1\pi^+ Np$ ($N > 0$) cross section from T2K on C_8H_8 [290] (right) as a function of reconstructed initial state nucleon momentum, $d\sigma/dp_N$, compared to current generator models. Dashed lines show the contributions that come from non-CCRPP processes, for example multi-pion and DIS interactions. The χ^2 shown in the legend indicate the level of agreement

NuWro challenges

Studies employing
NuWro 25.03 or 25.11

Karla Tame-Narvaez, Steven
Gardiner, Aleksandra
Ciprijanovic, Giuseppe
Cerati, arXiv 2603.09778

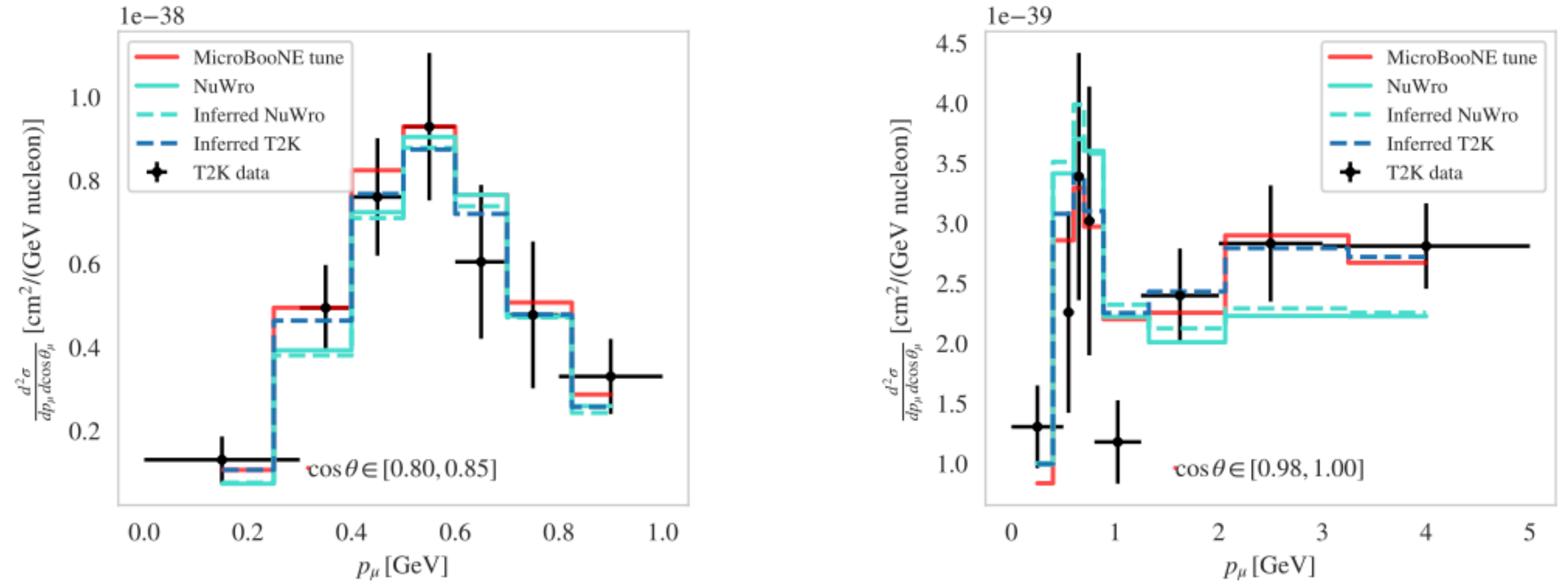


FIG. 6: Comparison of the T2K cross-section data (black points) to the NuWro [56] (solid cyan), NPE-inferred NuWro (dashed cyan), MicroBooNE Tune [9] (solid red), and NPE-inferred T2K (dashed blue) model predictions in two different slices of $\cos\theta$.

NPE - Neural Posterior Estimation

Final comments

- Since the very beginning NuWro has been providing an environment for training young people to become experts on neutrino interactions
- High quality of Monte Carlo generators is essential for a success of new neutrino oscillation experiments: HyperKamiokande and DUNE
- NuWro has been used many times as a laboratory to develop and/or investigate new models
- Monte Carlo generators are necessary tools to benchmark new theoretical models
- It is because all recent cross sections measurements are inclusive and include final state interactions (FSI) effects

I hope, we will have three days of fruitful discussions!

Thank you!